

Project Summary: AM-Based Communication System for Aircraft

This project investigates the use of Double Side Band Full Carrier – Amplitude Modulation (DSBFC-AM) for airport voice and text communications. The approach is justified by AM's long-distance propagation and spectral efficiency compared to FM.

Key Objectives:

- Enable clear voice and data communication with aircraft
- Determine bandwidth requirements and channel spacing
- Scale the system to meet operational and revenue goals
- Choose appropriate analog communication hardware

Technical Overview:

- 7 channels initially (for 7 aircraft), expanding to 14
- 25 kHz channel spacing over 175–350 kHz total bandwidth
- Use of components like LM3886 (audio amplifier), LMX2694-EP (VCO), LTC6431-20 (IF amp), and 74HC4067 (FDM unit)

Highlights:

- Revenue modeling using takeoff and landing charges
- Efficient use of spectrum through FDM and AM
- Design allows for cost-effective scalability and modularity

Visual Aids:

Below are relevant diagrams representing communication system components and bandwidth requirements.

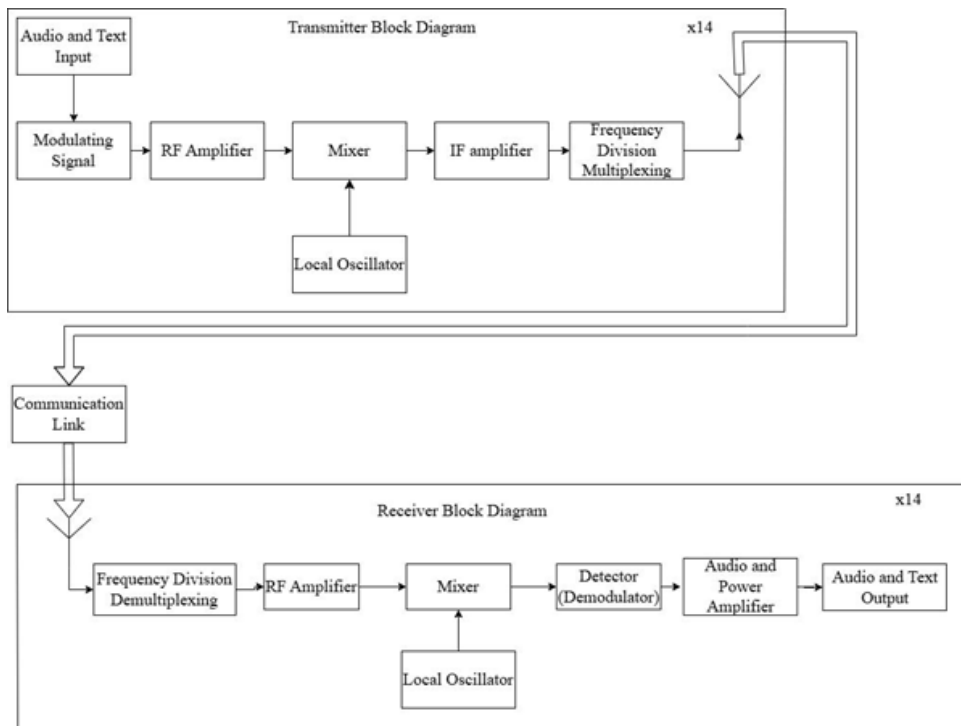


Figure 1: Initial Transmitter and Receiver Block Diagram –

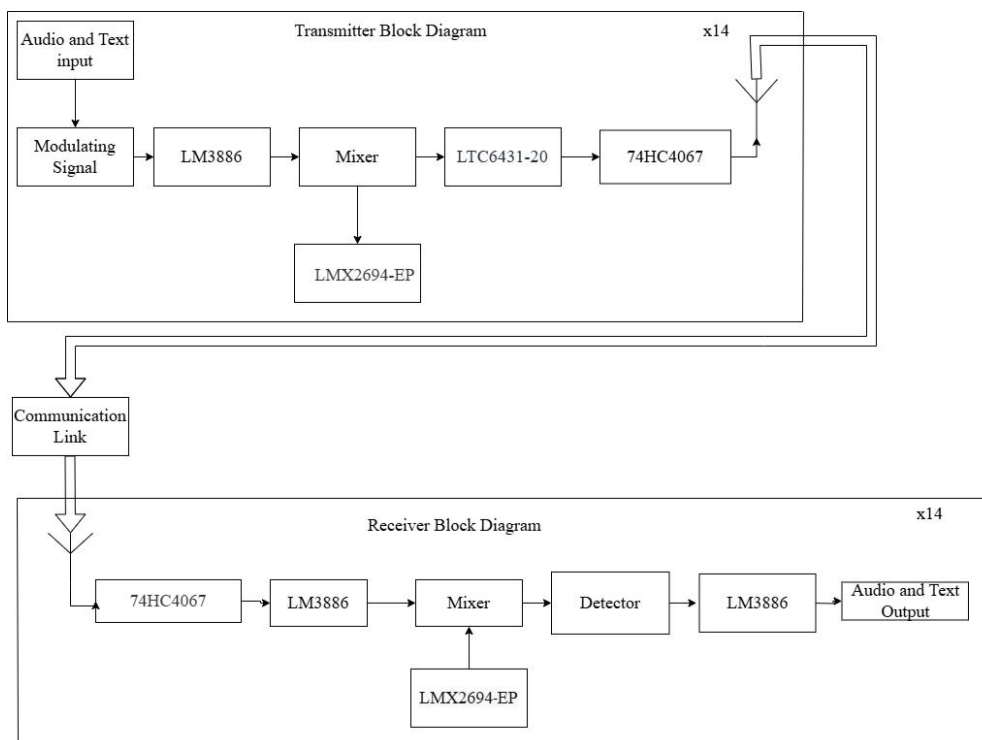


Figure 2: Transmitter and Receiver Block Diagram with the implementation of IC's

Conclusion:

DSBFC-AM is a practical and efficient solution for aviation communication due to its long-range capability, ease of implementation, and compatibility with analog multiplexing systems. Hardware choices ensure flexibility and signal integrity.